

Data Cleaning & Preparation

Project 1

Intern Name:	Maleeha Fatima	Date:	20 th may, 2026	Tool Used:	Excel
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Phase 1 — Missing Value Treatment (Strategic Imputation)

Change ID	Column Name	Issue Found	Method Used	Description of Action	Rows Preserved	Status
CR001	CouponCode	309 missing values	Mode Imputation (Structural Placeholder)	Identified missing values using =COUNTBLANK() and bulk-filled them with the placeholder string "NONE".	Preserved data completeness of rows without deleting records.	Resolved

Phase 2 — Duplicate Removal (Integrity Audit)

Change ID	Column Name	Duplicates Found	Method Used	Description of Action	Rows Preserved	Status
CR002	OrderID	0 duplicates	Deduplication Audit	Ran the Remove Duplicates tool on the Primary Key column.	Found 0 duplicates; verified a perfect 0% error rate for transaction IDs.	Resolved

Phase 3 — Data Format Standardization

Change ID	Column Name	Issue	Method Used	Description of Action	Impact	Status
CR003	Date	Date formats	ISO 8601 (YYYY-MM-DD)	Applied uniform parsing to convert variant strings into standard database timestamps (yyyy-mm-dd).	Achieved 0% date error threshold; alignment ensures accurate timeline sorting across reporting cycles.	Resolved
CR004	Product, ShippingAddress, PaymentMethod, OrderStatus, ReferralSource	Data Inconformity & Mixed Casing	Nested Function Normalization (PROPER + TRIM)	Stripped all invisible trailing spaces and converted mixed structural casing into uniform text titles.	Standardized categorical value tracking; prevents downstream aggregation discrepancies in pivot tables.	Resolved

Change ID	Column Name	Issue	Method Used	Description of Action	Impact	Status
CR005	TrackingNumber, CouponCode	Trailing Whitespaces (Data Noise)	String Whitespace Cleanse (TRIM Only)	Purged trailing and leading whitespaces from alphanumeric keys while preserving literal character case structures.	Secured structural code lookups; prevents false non-matches during query filtering.	Resolved
CR006	UnitPrice, TotalPrice	Truncated Decimals / Varying Precision	Fixed Numeric Precision Standard	Forced formatting configuration constraints across currency rows to display exactly 2 decimal units.	Brings pricing parameters into alignment with standard professional accounting and financial reporting rules.	Resolved